

Sleep & Wake-up Times

How long the MCU and the NFC reader take to go to sleep and to wake up.

STM32L432KB MCU · Stop 2 between events

Step	Time	Source
GOING TO SLEEP		
Arm LPTIM1 for the wake-up time (compare write)	≈ 92 μs 3 LSE cycles	firmware
Shortest sleep taken in Stop 2 (shorter: Sleep mode)	4 ms	firmware
Longest single Stop 2 sleep (loop caps it at 5 s)	15 s	firmware
Wake-up resolution (1 LPTIM tick, rounded up)	244 μs	firmware
WAKING UP		
Sleep → Run	6 cycles 1.5 μs	datasheet
Stop 2 → Run, MSI 4 MHz, from flash	12.06 μs 13.16 max	datasheet
MSI restart (inside the Stop 2 figure)	3 μs 6 max	datasheet
Standby → Run (power button), then reset	19.1 μs 25.8 max	datasheet
Power-on reset (battery connected)	250 μs 400 max	datasheet
HSI48 start-up (USB attach)	2.5 μs 6 max	datasheet
LSE crystal start-up (battery connected)	≈ 2 s	datasheet

ST25R3916 NFC reader · own wake-up timer, every 100 ms

Step	Time	Source
GOING TO SLEEP		
Field off → re-arm measurement	392 μs	measured
Measure amplitude (new reference)	25 μs max	datasheet
Wake-up check period (its own timer)	100 ms	firmware
Re-arm → first wake-up check	98.8 ms	measured
WAKING UP		
One wake-up check: oscillator start → field	605 μs	measured
One wake-up check: whole event (field 41 μs)	731 μs	measured
IRQ → oscillator restart (MCU over SPI)	1.27 ms	measured
Oscillator restart → field on	1.22 ms	measured
Card seen → field on	2.49 ms	measured
Field guard before the first command	5 ms + 1 tick	firmware
Field on: card read / empty poll	12.2 / 5.7 ms	measured

Datasheets: STM32L432 DS11451, ST25R3916 DS12484. Measured: scope captures TEK00005-07 across a 1 Ω shunt, 8 October 2026. The MCU wakes in ~12 μs; the reader's ~2.5 ms from IRQ to field is crystal start-up and SPI at 500 kHz.